

How a three-signal model flags failing companies early

Kestrel Bank — Business Loan Risk Model

Kestrel Bank, a UK business lender, currently only acts after a missed payment — by which point the borrower may already be insolvent and the loan at risk of not being repaid. Using public Companies House records for 9,844 UK companies (Ltd, PLC and LLP), three indicators — director resignations, overdue filings, and outstanding charges — were tested against known outcomes (Formal insolvency) to see how reliably they flag warning signals before it reaches that point. Combined into a single 0–3 risk score, the signals cleanly separate failing companies from healthy ones, and each additional signal sharply cuts false alarms.

Model Performance by Warning-Sign Threshold

Warning Signs	Failing companies detected by the model	Healthy companies wrongly flagged by the model
1	69.6%	24.2%
2	20.1%	2.3%
3	2.1%	0.1%

Detection and false-positive rates both fall as the threshold rises — by design. One sign catches broadly for review; three isolates only the clearest cases for escalation.

Recommendations

01 Warning Sign — flag for closer review before loan approval, not automatic rejection. Catches 69.6% of companies that go on to fail; roughly 1 in 4 healthy companies is wrongly flagged.

02 Warning Signs — escalate to a mid-term audit of the existing loan. Wrongly flagged healthy companies drop to 2.3% while still catching 1 in 5 failures.

03 Warning Signs — escalate to a senior underwriter for rejection review. Nearly noise-free, with only 0.1% of healthy companies wrongly flagged, though too narrow a net to rely on alone.

Built on a balanced sample of failed and active companies, not real-world default rates — these figures show how well the model discriminates between the two groups, not the odds that a flagged company actually fails. The model informs lending decisions; it does not automate them.